UNIVERSITY OF ENGINEERING AND TECHNOLOGY PESHAWAR

NOWSHERA CAMPUS

DEPARTMENT OF ELECTRICAL ENGINEERING

SMART SCHOOL MANAGEMENT SYSTEM

FINAL PROJECT REPORT

Course Title	Microprocessor Based System Design (EE 326)
Project Type	Embedded System and School Automation
Submitted To	Dr. Irfan Ahmad
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We are also very thankful to our respected teacher, Dr. Irfan Ahmad, for his guidance and support. His instructions helped us improve our idea and complete the project in a proper way.

We also thank our department and university for giving us this chance to work on a practical embedded system project. This project helped us understand microcontrollers, simulation, hardware design, and real school problems.

We are also thankful to the schools, teachers, and students who helped us during the survey. Their answers helped us choose a useful project.

ABSTRACT

The Smart School Management System is an embedded system project based on the AT89S51 microcontroller. The aim of this project is to solve common school management problems using a simple and low cost electronic system.

Before selecting the final idea, a survey was conducted in seven schools of the Paharpur region. The survey showed that many schools still use manual systems for attendance, bell timing, teacher monitoring, and emergency alerts. These manual methods waste time and can create mistakes.

Our project has four main modules. These are an automatic bell system, a live teacher and class mapping system, an emergency alert system, and a smart student attendance system using a 4 by 3 keypad.

The system uses a DS1307 real time clock for accurate time. It uses two LCDs. A 20 by 4 LCD is used for the main hall display. A 16 by 2 LCD is used for the principal office display. A keypad is used to enter student IDs for attendance. Push buttons are used for class teacher arrival and emergency alert. A buzzer is used for bell and emergency alarm.

The project was first designed and tested in Proteus 8 Professional. The code was written in Keil uVision 5 using Embedded C for the 8051 microcontroller. After simulation, the system was implemented on hardware.

This project is simple, useful, low cost, and easy to upgrade in the future.

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1. INTRODUCTION

1.1 Project Overview

The Smart School Management System is a microcontroller based project. It is designed to help schools manage daily work in a better way.

Many schools still depend on manual systems. A person rings the bell. Teachers take attendance on paper. The principal may not know if a teacher is late to class. Emergency alerts are also not fast in many schools.

Our project solves these problems with one embedded system. It uses an AT89S51 microcontroller as the main controller. It controls the bell, LCD displays, keypad attendance, teacher arrival buttons, and emergency buzzer.

The system is designed for schools in rural and semi urban areas. It is low cost and easy to use.

1.2 Background

Schools need good time management and discipline. A few minutes lost in each period can waste many hours over a full month. Manual attendance also takes time. Paper records can be lost or damaged.

During our survey, teachers and students shared many problems. The most common problems were manual attendance, late bell ringing, weak communication, and lack of quick emergency alert.

This project was selected to solve these real problems.

1.3 Need of the Project

This project is needed because schools need a simple system that can:

- Ring the bell automatically + Show lecture and teacher information
- Inform the principal when a teacher is missing
- Record student attendance through keypad
- Give emergency alert quickly
- Reduce manual work + Save time during school routine

1.4 Scope of the Project

The project covers four main school management tasks.

Automatic Bell System

The system rings the bell at fixed lecture times.

Teacher Class Mapping System

The main LCD shows which teacher is assigned to the class.

Principal Office Alert System

The principal LCD shows if a teacher is waiting, present, or missing.

Keypad Attendance System

Students enter their three digit ID and attendance is saved.

Emergency Alert System

An emergency button turns on the buzzer and shows warning messages on both LCDs.

2. SURVEY AND PROBLEM STUDY

2.1 Survey Purpose

Before starting the project, we conducted a survey. The purpose was to understand the real problems faced by schools.

We did not want to choose a random idea. We wanted a project that can solve real issues. So we visited different schools and collected information from students and teachers.

2.2 Survey Area

The survey was conducted in the Paharpur region of Pakistan. Seven schools were included in the survey. Both government and private schools were selected.

2.3 Schools Surveyed

No	School Name	Type	Location
1	Government Higher Secondary School Lar	Government	Lar
2	Government Primary School Lar	Government	Lar
3	Sadaqat Public School Lar	Private	Lar
4	Pak Pearl School Rangpur	Private	Rangpur
5	Al Abbas Science School and College	Private	Bilot
6	Government High School Kacha Mali Kheil	Government	Kacha Mali Kheil
7	Garrison High School	Semi Government	Kirri Khesor

2.4 Survey Method

A structured questionnaire was used. It had 25 questions. The questions were written in English and Urdu. This helped all respondents understand the questions easily.

The survey included questions about:

- Attendance system
- Electricity use
- Bell timing
- Teacher availability
- Emergency safety
- Student distraction
- School transport
- Technology needs

2.5 Respondents

The respondents were students and teachers. This gave us both views. Students shared their daily class problems. Teachers shared management and record keeping problems.

2.6 Key Survey Findings

The survey gave useful results. The main problems were:

Manual Attendance

Attendance is still taken on paper. It takes time. Records can be lost.

Manual Bell System

In many schools, a person rings the bell. Sometimes the bell is late. This affects the timetable.

Teacher Delay in Class

Sometimes a teacher does not reach the class on time. The principal may not know this quickly.

Weak Emergency Alert

Many schools do not have a proper emergency alarm system. This can be risky during fire, medical issue, or security danger.

Paper Based Records

School records are kept on paper. These records are hard to manage.

Technology Demand

Many respondents suggested smart attendance, automatic bell, and better school display systems.

2.7 Suggestions from Respondents

The most common suggestions were:

- Automatic attendance system
- Automatic bell system
- Smart display board
- Better teacher monitoring
- Emergency siren
- Digital school records
- Homework and learning support tools

2.8 Survey Conclusion

The survey helped us select a useful project. It showed that schools need simple automation. The problems are real. The solution should be low cost, simple, and reliable.

Our final project was selected after studying these survey results. 3. IDEA SELECTION

3.1 Initial Ideas

After the survey, many ideas were discussed. Some of them were:

- Smart attendance system
- Automatic bell system
- Smart whiteboard
- Digital result system
- School transport tracking
- Emergency alert system
- Complaint management system
- Teacher monitoring system

3.2 Final Selected Idea

After discussion, we selected the Smart School Management System. This idea includes four useful modules in one project.

These modules are:

- Automatic bell system
- Live teacher class mapping system
- Emergency alert system
- Smart student attendance system

3.3 Reasons for Selecting This Idea

It Solves Real Problems

The selected idea is based on survey results. It solves the problems that schools face every day.

It Is Possible With Our Course Knowledge

The project can be made using an 8051 microcontroller and Embedded C. This matches our course.

It Is Low Cost

The components are easily available. The system does not need expensive sensors or internet for the basic version.

It Is Easy to Demonstrate

All modules can be shown clearly in simulation and hardware.

It Has Future Scope

The same system can be upgraded with mobile app, database, and parent notification system.

It Helps Students and Teachers

The system saves time and improves school discipline.

3.4 Why Other Ideas Were Not Selected

Some ideas were useful but not selected for the first version.

Smart whiteboard needs expensive hardware. Transport tracking needs GPS, GSM, and more advanced security features. Face recognition and fingerprint systems need extra modules and more time.

So we selected a focused system that can be completed and tested properly.

3.5 Benefits of Selected Idea

The selected idea gives many benefits.

- Saves time in attendance
- Rings bell on exact time
- Reduces teacher absence problem
- Helps principal monitor classes
- Gives quick emergency alert
- Reduces manual work
- Improves school discipline
- Can be upgraded in future

4. OBJECTIVES OF THE PROJECT

The main objectives of the project are:

- To design a microcontroller based school management system
- To automate the school bell using real time clock
- To show time and lecture information on LCD
- To display teacher information for classes
- To alert the principal if a teacher is missing
- To record student attendance using keypad ID entry
- To detect duplicate attendance entry
- To show valid and invalid student ID messages
- To provide emergency alert using button and buzzer
- To simulate the complete system in Proteus 8 Professional
- To implement the working circuit on hardware

5. SYSTEM COMPONENTS

5.1 Main Components Used

Component	Purpose
AT89S51 Microcontroller	Main control unit
11.0592 MHz Crystal	Clock source for microcontroller
DS1307 RTC Module	Real time clock for time keeping
20 by 4 LCD	Main hall display
16 by 2 LCD	Principal office display
4 by 3 Matrix Keypad	Student ID entry
Push Buttons	Teacher arrival and emergency input
Buzzer	Bell and emergency siren
Resistors	Pull up and current control
Capacitors	Oscillator and reset circuit
5V DC Supply	Power for the system
Connecting Wires	Circuit connections
PCB or Breadboard	Hardware implementation
Relay Module	Optional for real bell or siren load

5.2 Software Tools Used

Software	Purpose
Keil uVision 5	Writing and compiling Embedded C code
C51 Compiler	Compiler for 8051 code
Proteus 8 Professional	Circuit simulation
Hex File	Program file loaded into microcontroller
Programmer Software	Loading code into AT89S51 hardware

5.3 Microcontroller Used

The main controller is AT89S51. It is an 8051 family microcontroller. It is used because it is simple, low cost, and suitable for learning embedded systems.

The AT89S51 controls all inputs and outputs. It reads the keypad and buttons. It communicates with the RTC. It sends data to both LCDs. It also controls the buzzer.

5.4 Real Time Clock

The DS1307 real time clock is used for accurate time. It keeps track of hours, minutes, and seconds.

The microcontroller reads time from the RTC using software I2C communication. The RTC helps the system ring the bell at correct lecture times.

5.5 LCD Displays

Two LCDs are used.

Main Hall LCD

A 20 by 4 LCD is used in the main hall. It shows time, welcome message, lecture number, next bell time, teacher names, attendance messages, and emergency warning.

Principal Office LCD

A 16 by 2 LCD is used in the principal office. It shows the status of class A and class B. It also shows attendance result and emergency alert.

5.6 Keypad

A 4 by 3 matrix keypad is used for student attendance. Students enter a three digit ID. In this project, valid demo IDs are 101 to 110.

The keypad layout is:

1 2 3

4 5 6

7 8 9

* 0 #

The star key is used to enter attendance mode. The hash key is used to save attendance.

5.7 Push Buttons

Three push buttons are used.

- Class A teacher arrival button
- Class B teacher arrival button
- Emergency button

The Class A and Class B buttons are used to mark teacher arrival. The emergency button is connected to INT0 of the microcontroller.

5.8 Buzzer

The buzzer works as a bell and emergency siren. In the existing circuit, the buzzer is active low.

This means:

- Logic 0 turns buzzer ON
- Logic 1 turns buzzer OFF

5.9 Power Supply

A regulated 5V DC supply is used. The microcontroller, RTC, LCDs, keypad, and other parts work on 5V.

Port 0 pins need external pull up resistors because Port 0 of 8051 is open drain.

6. SYSTEM DESIGN

6.1 Block Diagram Description

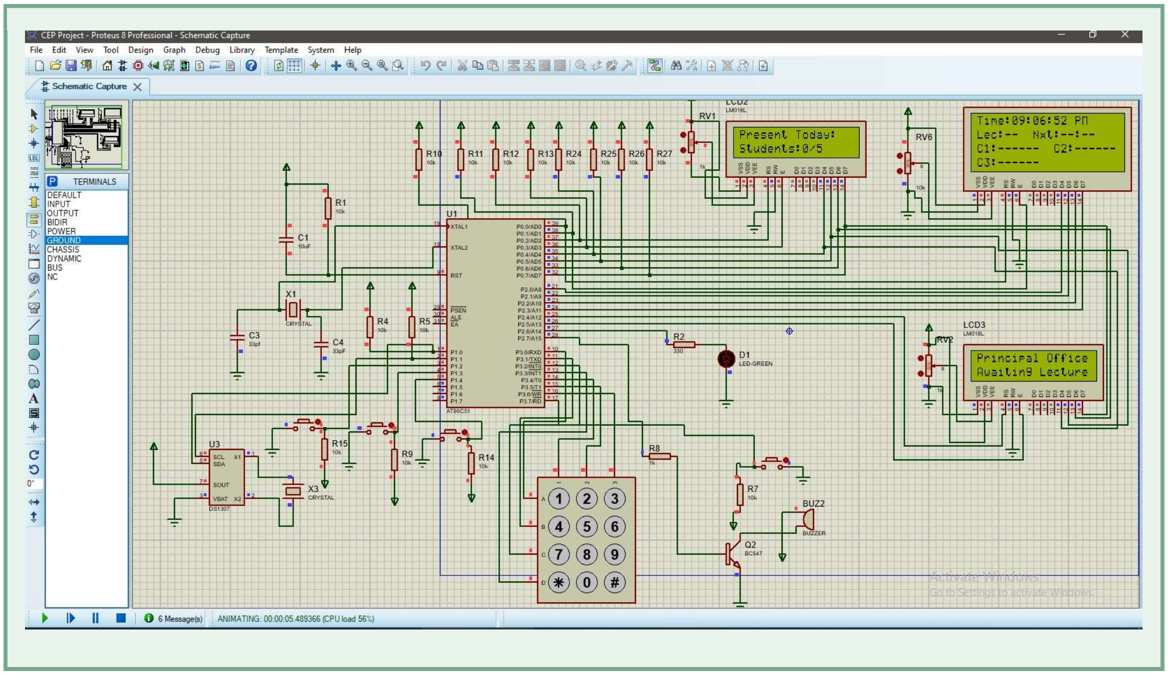
The system has one main microcontroller. All modules are connected with it.

The RTC gives real time to the microcontroller. The microcontroller checks the lecture schedule. When the time matches, it turns on the buzzer and updates the LCDs.

The keypad sends student ID input. The microcontroller checks if the ID is valid. It then saves the attendance in memory.

Class buttons are used to confirm teacher arrival. If a teacher does not arrive within 5 minutes, the principal LCD shows missing status.

The emergency button starts emergency mode. Both LCDs show emergency message and the buzzer turns ON.



6.2 Final Port Allocation

Microcontroller Pin	Connected Device
P1.0	Keypad Row A
P1.1	Keypad Row B
P1.2	Keypad Row C
P1.3	Keypad Row D
P1.4	Keypad Column 1
P1.5	Keypad Column 2
P1.6	Keypad Column 3
P1.7	Free
P2.0 to P2.7	Principal Office LCD Data
P0.0	Main Hall LCD RS

Microcontroller Pin	Connected Device
P0.1	Main Hall LCD EN
P0.2	Principal LCD RS
P0.3	Principal LCD EN
P0.4	DS1307 SDA
P0.5	DS1307 SCL
P3.0	Class A Button
P3.1	Class B Button
P3.2	Emergency Button and INT0
P3.3	Buzzer
P3.4	Main Hall LCD D4
P3.5	Main Hall LCD D5
P3.6	Main Hall LCD D6
P3.7	Main Hall LCD D7

6.3 LCD Connection Design

The main hall LCD is connected in 4 bit mode. This saves microcontroller pins. Its data pins D4 to D7 are connected to P3.4 to P3.7.

The principal LCD is connected in 8 bit mode. Its data pins are connected to Port 2.

The RW pins of both LCDs are connected to ground. So the LCDs are used only in write mode.

6.4 RTC Connection Design

The DS1307 RTC uses two lines.

- SDA
- SCL

These are connected to P0.4 and P0.5. Since Port 0 needs pull up resistors, external pull ups are required on these pins.

6.5 Keypad Connection Design

The keypad has four rows and three columns. Rows are connected to P1.0 to P1.3. Columns are connected to P1.4 to P1.6.

The code scans one row at a time. If any column becomes low, the pressed key is detected.

6.6 Button Connection Design

The Class A and Class B buttons are connected to P3.0 and P3.1. These buttons are active low.

The emergency button is connected to P3.2. This pin is also INT0. It gives a fast interrupt when emergency is pressed.

7. DETAILED WORKING OF THE PROJECT

7.1 Overall Working

When the system starts, all ports are initialized. Both LCDs are initialized. Timer 0 and external interrupt 0 are enabled.

The RTC is read. The main hall LCD shows the current time and welcome message. The principal LCD shows that there is no lecture at the start.

The system then keeps running in a continuous loop. It reads time, checks lecture schedule, scans buttons, scans keypad, updates LCDs, and handles emergency.

7.2 Automatic Bell System

Main Function

The automatic bell system rings the buzzer at fixed lecture times.

In the code, three lecture times are stored:

Lecture	Time
Lecture 1	08:00
Lecture 2	08:10
Lecture 3	08:20

The microcontroller reads time from the DS1307 RTC. It compares the current hour and minute with stored lecture times.

When the time matches, the system starts a new lecture. It updates the lecture number and teacher names on the main hall LCD. It also rings the buzzer for bell indication.

The code uses a done array. This prevents the bell from ringing again and again during the same minute.

Benefit

This removes the need for a person to ring the bell manually. The bell rings on exact time.

7.3 Teacher Class Mapping System

Main Function

This module shows which teacher is assigned to the class.

The main hall LCD shows lecture information. It also shows teacher names for Class A and Class B.

In the code, teacher names are shown as:

- Teacher A
- Teacher B
- Teacher C
- Teacher D
- Teacher E
- Teacher F

For each lecture, two teachers are assigned. One teacher is for Class A and one teacher is for Class B.

Example:

Lecture 1

Class A: Teacher A

Class B: Teacher B

Lecture 2

Class A: Teacher C

Class B: Teacher D

Lecture 3

Class A: Teacher E

Class B: Teacher F

Benefit

Students and staff can easily see which teacher should be in which class.

7.4 Principal Office Monitoring System

Main Function

This module tells the principal if the teacher is present, waiting, or missing.

When a new lecture starts, the status of Class A and Class B is set to waiting. A timer starts for both classes.

If the teacher reaches Class A, the Class A button is pressed. The system marks Class A teacher as present.

If the teacher reaches Class B, the Class B button is pressed. The system marks Class B teacher as present.

If any teacher does not arrive within 5 minutes, the system marks that teacher as missing. The principal LCD then shows missing status.

The Timer 0 interrupt is used for this delay. It counts time in the background. After 300 seconds, the missing alert is generated.

Benefit

The principal can know the class status without visiting each classroom.

7.5 Emergency Alert System

Main Function

This module gives fast warning during danger.

The emergency button is connected to INT0. When it is pressed, the interrupt runs immediately. The buzzer turns ON and emergency mode starts.

Both LCDs show emergency messages.

Main hall LCD shows:

!! EMERGENCY !!

EVACUATE NOW!

Press BTN to cancel

Principal LCD also shows emergency alert.

When the emergency button is pressed again after release, the emergency mode is cancelled. The buzzer turns OFF and the normal screen is restored.

Benefit

This gives a fast warning in case of fire, medical emergency, or security danger.

7.6 Smart Student Attendance System

Main Function

This module records student attendance using keypad ID entry.

A student presses star to enter attendance mode. The main hall LCD shows attendance screen. The student enters a three digit ID. Then the student presses hash to save.

Valid demo IDs are:

101, 102, 103, 104, 105, 106, 107, 108, 109, 110

If the ID is valid and not already marked, attendance is saved. The LCD shows attendance saved. The buzzer gives a short beep.

If the same ID is entered again, the system shows already present.

If the ID is outside the valid range, the system shows invalid ID.

The attendance is stored using bits inside an integer variable. Each student has one bit. This saves memory.

Keypad Controls

Key	Function
*	Enter attendance mode
Digits	Enter student ID
#	Save attendance
* during entry	Clear last digit
# outside mode	Show attendance summary

Benefit

This removes manual attendance. It is faster and reduces errors.

7.7 Attendance Summary

When hash is pressed outside attendance mode, the principal LCD shows attendance summary.

It shows:

Present students out of total students.

For example:

Present 05/10

This helps the administration see total attendance quickly.

7.8 RTC Working

The DS1307 stores time in BCD format. The code converts BCD to decimal and decimal to BCD.

The RTC is read again and again in the main loop. The current time is shown on the main hall LCD.

The RTC also controls the bell schedule.

7.9 Timer 0 Working

Timer 0 is used in mode 1. It creates a delay of about 50 milliseconds. After 20 counts, about 1 second is completed.

This one second count is used to check the 5 minute teacher arrival time.

If 300 seconds pass and the teacher is not marked present, the status becomes missing.

7.10 Interrupt Working

External interrupt INT0 is used for emergency. This is important because emergency must be fast.

The interrupt only changes flags and buzzer status. LCD work is not done inside the interrupt. This is a good programming method because LCD functions take time.

The main loop reads the emergency flag and then updates LCDs.

7.11 Buzzer Working

The buzzer is used in three places.

- Bell ringing
- Attendance saved beep
- Emergency alarm

The buzzer in our circuit is active low. So the code turns it ON by sending logic 0 and turns it OFF by sending logic 1.

7.12 Normal Display Working

In normal mode, the main LCD shows time on the first line. It also shows welcome message or lecture information.

Before lecture starts, it shows:

Time

Welcome to SSMS

Next Bell

Press star Attendance

During lecture, it shows lecture number, next lecture time, and teacher names.

7.13 Principal LCD Working

The principal LCD shows class status.

At the start, it shows no lecture.

During lecture, it can show:

- Waiting
- Teacher OK
- Missing

It also shows attendance result and emergency message when needed.

8. SOFTWARE DEVELOPMENT IN KEIL UVISION 5

8.1 Software Platform

The code was written in Keil uVision 5. The compiler used was C51. The programming language was Embedded C.

The target microcontroller was AT89S51.

8.2 Main Code Sections

The code is divided into different parts.

Delay Functions

These functions create millisecond and microsecond delays.

I2C Functions

These functions communicate with DS1307 RTC.

RTC Functions

These functions set and read time.

LCD Functions

These functions send commands and data to both LCDs.

Keypad Functions

These functions scan the keypad and detect pressed key.

Attendance Functions

These functions manage ID entry, save attendance, reject duplicate entry, and show summary.

Bell Function

This function turns the buzzer ON for bell duration.

Timer Interrupt

This checks teacher absence delay.

Emergency Interrupt

This handles emergency button fast.

Main Function

This initializes the system and runs the main control loop.

8.3 Hex File Generation

After writing the code, the project was built in Keil. The compiler checked the code and generated the hex file.

The hex file was then used in Proteus simulation. It was also used for hardware programming.

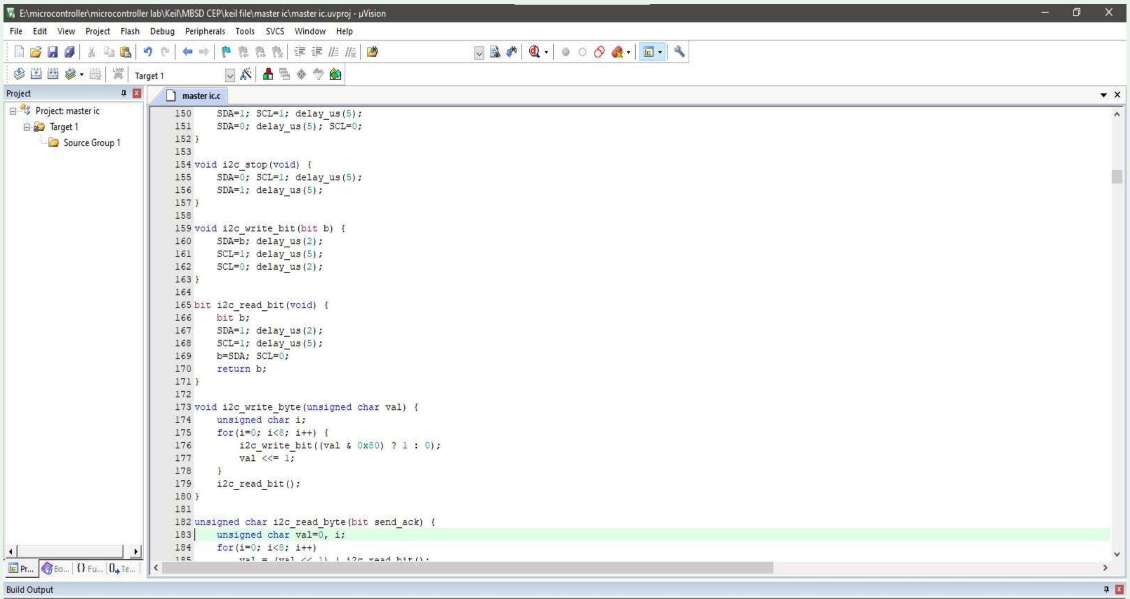


Figure 1:Keil Interface

Figure 2 : Built Code Successfully

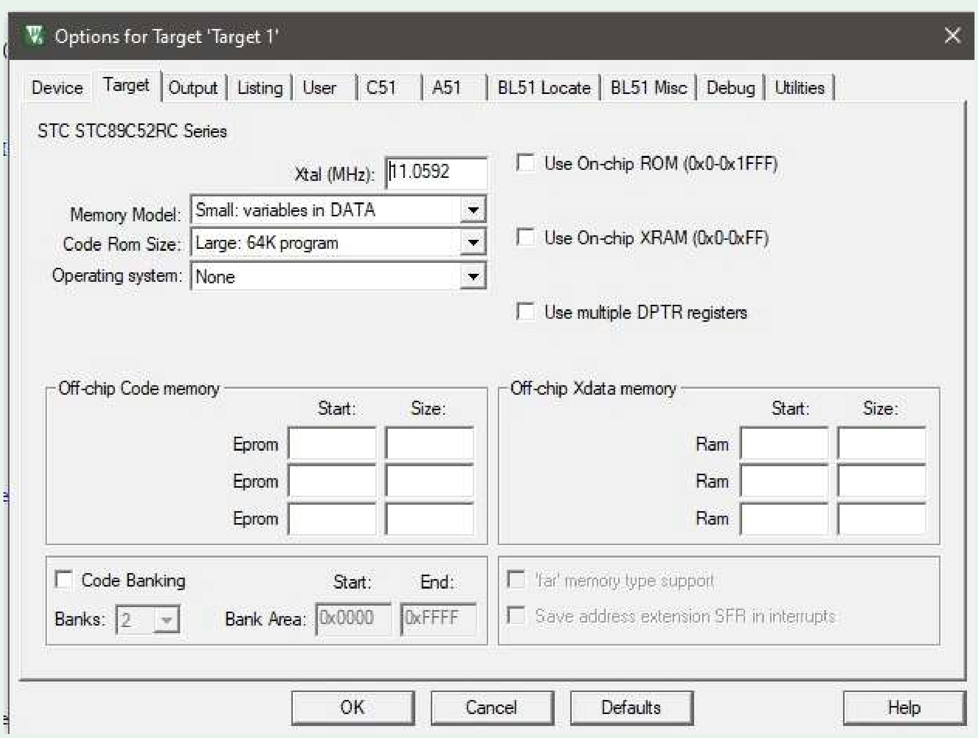


Figure 3: Creating Hex file

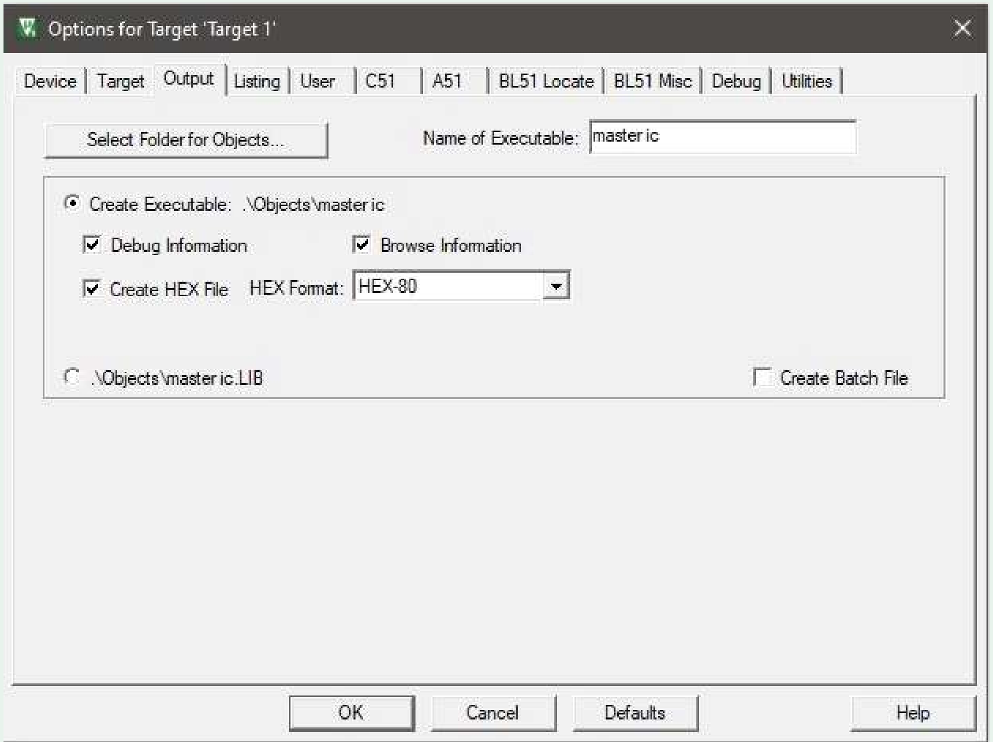


Figure 4: Creating Hex file

8.4 Important Software Settings

Setting	Value
Microcontroller	AT89S51
Compiler	Keil C51
Crystal Frequency	11.0592 MHz
Timer 0 Mode	Mode 1
RTC	DS1307
LCD 1	20 by 4 in 4 bit mode
LCD 2	16 by 2 in 8 bit mode
Attendance IDs	101 to 110
Emergency Input	INT0

9. SIMULATION IN PROTEUS 8 PROFESSIONAL

9.1 Purpose of Simulation

Before making the hardware, the complete system was tested in Proteus 8 Professional. Simulation helped us check the circuit without damaging real components.

It also helped us test the code, LCD display, keypad, RTC, buttons, and buzzer.

9.2 Proteus Components Used

The following components were used in Proteus:

- AT89S51 microcontroller
- DS1307 RTC
- 20 by 4 LCD
- 16 by 2 LCD
- 4 by 3 keypad
- Push buttons
- Buzzer
- Crystal oscillator
- Capacitors
- Resistors
- 5V supply
- Ground terminals

9.3 Simulation Circuit Design

The AT89S51 was placed at the center of the circuit. The RTC was connected to Port 0 pins through SDA and SCL lines. The main hall LCD was connected in 4 bit mode. The principal LCD was connected with Port 2 in 8 bit mode.

The keypad was connected to Port 1. The push buttons were connected to Port 3. The buzzer was connected to P3.3.

The hex file generated from Keil was loaded into the AT89S51 in Proteus.

9.4 Simulation Steps

The simulation was completed in these steps:

1. Designed the circuit in Proteus
2. Connected the LCDs, RTC, keypad, buttons, and buzzer
3. Generated the hex file from Keil
4. Loaded the hex file into AT89S51
5. Set the crystal frequency to 11.0592 MHz
6. Started the simulation
7. Checked the LCD time display
8. Tested bell timing
9. Tested teacher arrival buttons
10. Tested keypad attendance

11. Tested invalid and duplicate IDs
12. Tested emergency button

9.5 Simulation Testing

Time Display Test

The RTC time was shown correctly on the main hall LCD.

Bell Test

The buzzer turned ON at lecture start time.

Teacher Button Test

When Class A or Class B button was pressed, the principal LCD showed teacher OK.

Missing Teacher Test

If the button was not pressed for 5 minutes, the principal LCD showed missing.

Attendance Test

Valid student IDs were saved. Duplicate IDs were rejected. Invalid IDs were also rejected.

Emergency Test

When emergency button was pressed, both LCDs showed emergency message and buzzer turned ON.

9.6 Importance of Simulation

Simulation saved time and reduced mistakes. It helped us find connection issues before hardware. It also helped us confirm that the code logic was correct.

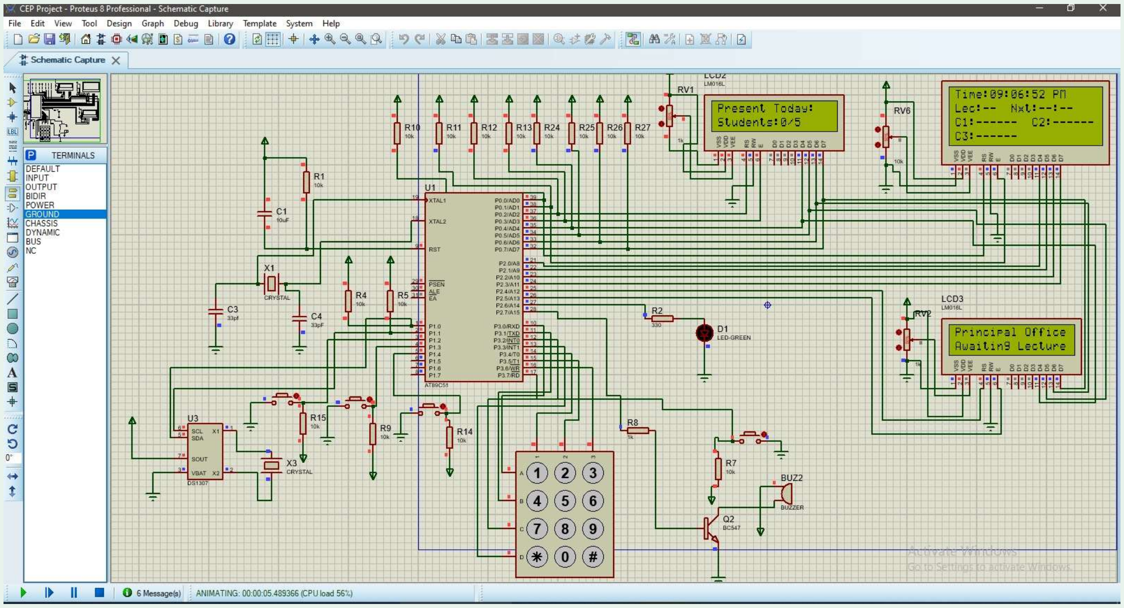


Figure 5: Complete Proteus Schematic

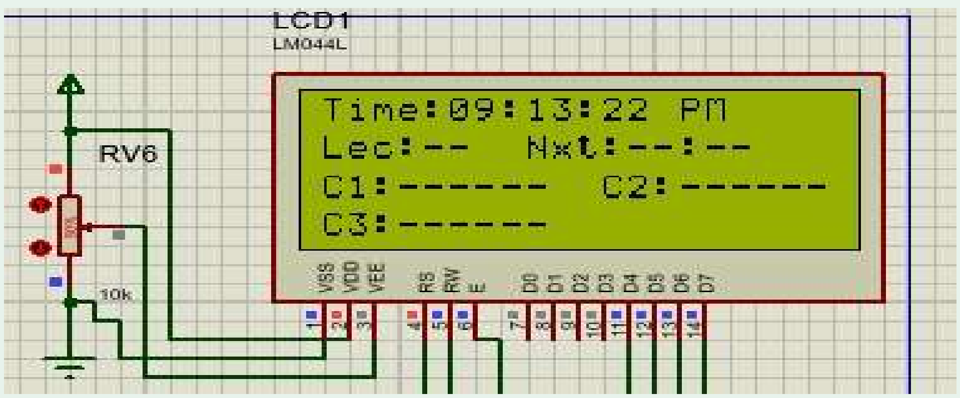


Figure 6: Main Hall LCD

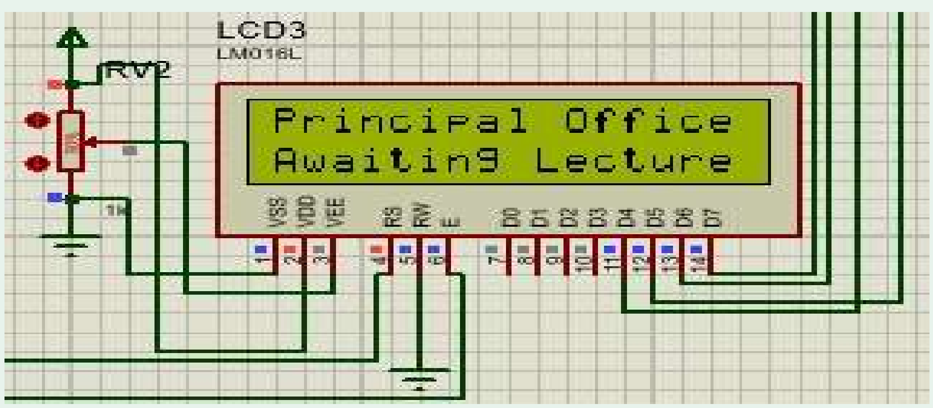


Figure 7:Principal Office LCD

Figure 8: Attendance LCD

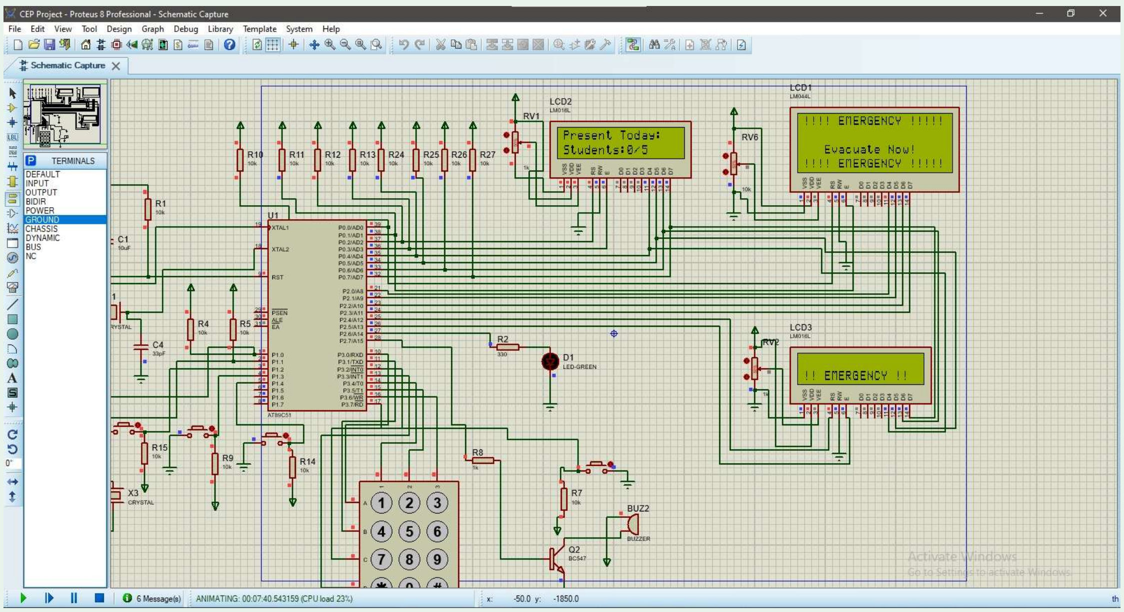


Figure 9: Emergency Alert ON

10. HARDWARE IMPLEMENTATION

10.1 Hardware Build Overview

After successful simulation, the circuit was implemented on hardware. The main goal was to make the same system work on real components.

The hardware included the AT89S51 microcontroller, crystal circuit, reset circuit, LCDs, RTC, keypad, buttons, buzzer, and power supply.

10.2 Power Supply

A regulated 5V DC supply was used. Proper 5V is very important for the microcontroller and LCDs.

The ground of all modules was connected together. This made the circuit stable.

10.3 Microcontroller Circuit

The AT89S51 was connected with an 11.0592 MHz crystal. Two capacitors were used with the crystal. A reset circuit was also connected to the reset pin.

This allowed the microcontroller to start properly when power was applied.

10.4 LCD Hardware Connection

The 20 by 4 LCD was connected in 4 bit mode. This reduced the number of pins used.

The 16 by 2 LCD was connected in 8 bit mode using Port 2.

The RW pins of both LCDs were connected to ground.

10.5 RTC Hardware Connection

The DS1307 RTC module was connected with SDA and SCL lines. External pull up resistors were used on Port 0 pins because Port 0 of 8051 needs pull ups.

The RTC module helped the system keep accurate time.

10.6 Keypad Hardware Connection

The keypad was connected to Port 1. Rows and columns were connected according to the code.

The keypad was tested by pressing each key and checking the LCD output.

10.7 Button Hardware Connection

Three buttons were used.

Class A button was connected to P3.0.

Class B button was connected to P3.1.

Emergency button was connected to P3.2.

The buttons were tested one by one.

10.8 Buzzer Hardware Connection

The buzzer was connected to P3.3. The circuit used an active low buzzer. So the buzzer turns ON when the microcontroller output is low.

For a larger school bell or siren, a relay driver can be used.

10.9 Hardware Testing Method

The hardware was tested step by step.

First, the microcontroller was tested. Then LCDs were tested. After that, RTC was tested. Then keypad and buttons were tested. Finally, the full system was tested together.

This step by step method helped us find faults easily.

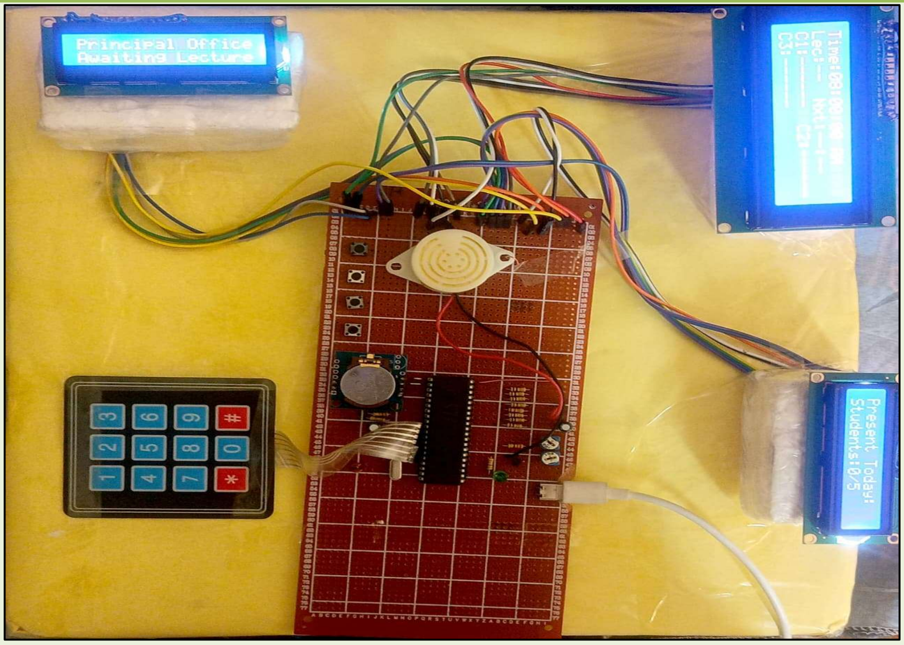


Figure 10:Full Hardware Overveiw

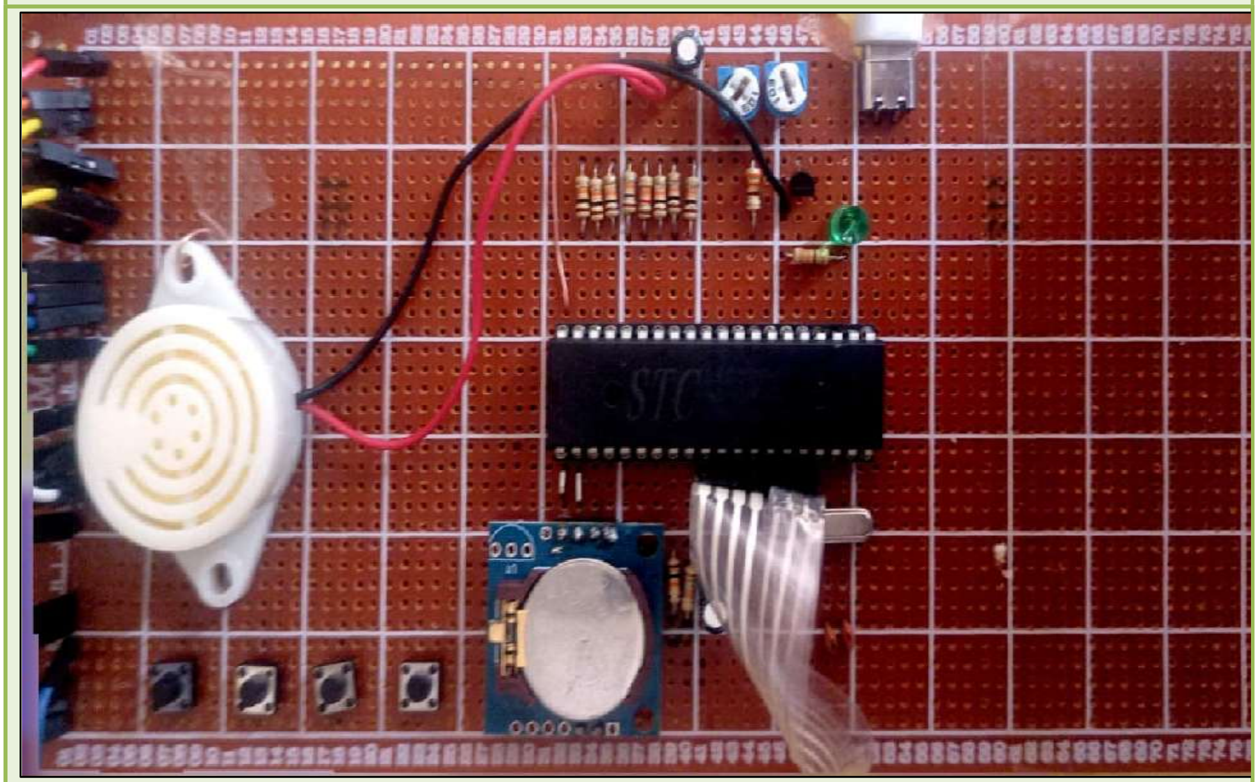


Figure 11: Vero board

Figure 12:Main Hall LCD

Figure 13:Principle office LCD

Figure 14:Attendence LCD

Figure 15:Keypad

11. TESTING AND RESULTS

11.1 Test Case 1: System Start

Input

Power was supplied to the circuit.

Expected Result

Both LCDs should initialize. Main LCD should show time and welcome message.

Actual Result

The LCDs started correctly. Time and welcome message were displayed.

Status

Successful.

11.2 Test Case 2: Automatic Bell

Input

RTC time reached lecture time.

Expected Result

Buzzer should ring and LCD should show lecture information.

Actual Result

The buzzer rang at the selected time. The LCD showed lecture number and teacher names.

Status

Successful.

11.3 Test Case 3: Teacher Arrival Button

Input

Class A or Class B button was pressed.

Expected Result

Principal LCD should show teacher OK.

Actual Result

The principal LCD showed teacher OK for the correct class.

Status

Successful.

11.4 Test Case 4: Missing Teacher Alert

Input

Teacher button was not pressed within 5 minutes.

Expected Result

Principal LCD should show missing.

Actual Result

The principal LCD showed missing after the timer completed.

Status

Successful.

11.5 Test Case 5: Valid Attendance ID

Input

Student entered ID 101 and pressed hash.

Expected Result

Attendance should be saved.

Actual Result

LCD showed attendance saved and buzzer beeped.

Status

Successful.

11.6 Test Case 6: Duplicate Attendance ID

Input

Student entered same ID again.

Expected Result

System should reject duplicate entry.

Actual Result

LCD showed already present.

11.7 Test Case 7: Invalid Attendance ID

Input

Student entered ID outside 101 to 110.

Expected Result

System should show invalid ID.

Actual Result

LCD showed invalid ID.

11.8 Test Case 8: Emergency Button

Input

Emergency button was pressed.

Expected Result

Buzzer should turn ON. Both LCDs should show emergency warning.

Actual Result

Buzzer turned ON and LCDs showed emergency message.

Status

Successful.

11.9 Final Result

The complete system worked successfully in simulation and hardware. All four modules were tested.

12. BENEFITS OF THE PROJECT

12.1 Time Saving

The automatic bell saves time. Keypad attendance also reduces time in class.

12.2 Better Discipline

The bell rings on time. Teacher status is shown clearly. This improves school discipline.

12.3 Improved Accountability

The principal can know if a teacher is missing. This reduces wasted lecture time.

12.4 Accurate Attendance

The keypad attendance system reduces mistakes. It also prevents duplicate attendance.

12.5 Better Safety

The emergency alert system gives quick warning. This can help during fire, medical issue, or other danger.

12.6 Low Cost

The system uses simple and available components. It is suitable for schools with limited budget.

12.7 Easy to Upgrade

The system can be upgraded with mobile app, digital records, and transport safety features.

13. FUTURE UPGRADATION

The project has a strong future scope. More features can be added later.

13.1 Mobile Application

A mobile application can be made for school administration. The principal or admin can change bell time, teacher names, and class schedule from a mobile phone.

This will make the system more flexible.

13.2 Digital School Record System

The system can be connected with a database. Student records, attendance history, exam results, and teacher records can be stored digitally.

This will reduce paper work.

13.3 Complaint Management System

A complaint system can be added. Students and teachers can submit complaints. The complaint can be sent to the correct staff member.

For example, a complaint about a fan can be sent to the electrician.

13.4 Parent Notification System

A transport safety system can be added. The school bus can use fingerprint, face recognition, GPS, or RFID. When a student enters or leaves the bus, parents can receive a message.

This will improve student safety.

13.5 RFID or Fingerprint Attendance

In future, keypad attendance can be replaced with RFID card or fingerprint module. This will make attendance faster.

13.6 Cloud Data Storage

Attendance and school data can be stored on cloud. The principal can check data from anywhere.

13.7 Energy Management

The system can be upgraded to control fans and lights. If a classroom is empty, fans and lights can turn OFF automatically.

13.8 Summary of Future Upgrades

Future Upgrade	Purpose	Priority
Mobile App	Control system from phone	Phase 2
Digital Records	Store school data digitally	Phase 2
Complaint System	Send complaints to correct staff	Phase 3
Bus Tracking	Improve student transport safety	Phase 3
RFID Attendance	Faster attendance system	Phase 3
Energy Control	Save electricity	Phase 3

14. PROBLEMS FACED DURING WORK

14.1 Port 0 Pull Up Issue

Port 0 of the 8051 does not have internal pull up resistors. At first, some signals were not stable. This was solved by using external pull up resistors on Port 0 pins.

14.2 LCD Display Issues

LCD timing is sensitive. Some characters were not shown correctly during early testing. Delay values were adjusted and the LCD worked properly.

14.3 Keypad Scanning

Keypad scanning needed proper row and column checking. Debouncing was also needed. A small delay was added after key press to avoid wrong input.

14.4 RTC Communication

The RTC uses I2C communication. The SDA and SCL lines need pull ups. After correct connections, the RTC worked.

14.5 Emergency Interrupt

The emergency button needed proper interrupt handling. LCD commands were not placed inside the interrupt because LCD functions take time. Flags were used instead.

14.6 Hardware Wiring

The circuit had many wires. Careful checking was needed to avoid wrong connections. Each module was tested separately before full testing.

15. CONCLUSION

The Smart School Management System was completed successfully. It was designed after a real survey of seven schools. The survey helped us understand actual problems in school management.

The final system includes automatic bell, teacher class mapping, principal office alert, keypad attendance, and emergency alert. The system uses AT89S51 microcontroller, DS1307 RTC, LCD displays, keypad, buttons, and buzzer.

The code was written in Keil uVision 5. The circuit was tested in Proteus 8 Professional. After simulation, the system was implemented on hardware.

The project is useful, low cost, and easy to understand. It can help schools save time, improve discipline, and increase safety. It also has many future upgrade options.

This project improved our understanding of microcontrollers, Embedded C, LCD interfacing, keypad scanning, RTC communication, timers, interrupts, simulation, and hardware implementation.

16. REFERENCES

1. AT89S51 Microcontroller Datasheet
2. DS1307 Real Time Clock Datasheet
3. Keil uVision 5 User Guide
4. Proteus 8 Professional Simulation Tool
5. 8051 Microcontroller and Embedded Systems by Muhammad Ali Mazidi
6. Course lectures of Microprocessor Based System Design EE 326
7. Survey data collected from seven schools in Paharpur region

17. APPENDIX

17.1 Important Project Settings

Item	Detail
MCU	AT89S51
Compiler	Keil C51
Crystal	11.0592 MHz
RTC	DS1307
Main LCD	20 by 4
Principal LCD	16 by 2
Keypad	4 by 3
Valid Student IDs	101 to 110
Emergency Input	INT0
Buzzer Type	Active Low

17.2 Attendance Operation

1. Press star key to enter attendance mode.
2. Enter three digit student ID.
3. Press hash key to save.
4. If ID is valid, attendance is saved.
5. If ID is already saved, duplicate message is shown.
6. If ID is invalid, invalid ID message is shown.
7. Press hash outside attendance mode to see summary.

17.3 Lecture Schedule Used in Project

Lecture	Time	Class A Teacher	Class B Teacher
Lecture 1	08:00	Teacher A	Teacher B
Lecture 2	08:10	Teacher C	Teacher D
Lecture 3	08:20	Teacher E	Teacher F

Thank you